

Course 50.050/50.550

Advanced Algorithms

Week 4 – Cohort Class L04.03



Outline of Cohort Class

- ▶ Exercise on adjacency matrices
- ▶ Class activity: Matching puzzle
- ▶ Hall's marriage theorem
- ▶ Discussion on matching algorithms



Recap: Adjacency matrix

Definition: Let $G = (V, E)$ be a **simple** graph, and suppose that $V = \{v_1, \dots, v_n\}$. The **adjacency matrix** of G is an n -by- n square matrix $A = [a_{i,j}]$, whose (i,j) -th entry $a_{i,j}$ is defined as follows:

- ▶ (If G is undirected): $a_{i,j} = \begin{cases} 1, & \text{if } v_i \text{ is adjacent to } v_j; \\ 0, & \text{otherwise.} \end{cases}$
- ▶ (If G is directed): $a_{i,j} = \begin{cases} 1, & \text{if } (v_i, v_j) \in E; \\ 0, & \text{otherwise.} \end{cases}$

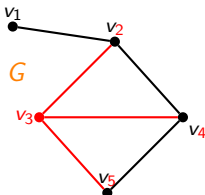
Basic properties:

- ▶ Since G has no loops, the diagonal entries $a_{i,i}$ of A are all zeros.
- ▶ The adjacency matrix of an **undirected** graph is symmetric.
 - ▶ $A = [a_{i,j}]$ is called **symmetric** if $a_{i,j} = a_{j,i}$ for all $1 \leq i, j \leq n$.
- ▶ For an undirected graph G , we have:
(row sum of i -th row) = (column sum of i -th column) = $\deg(v_i)$.
- ▶ For a directed graph G , we have:
 - ▶ The row sum of the i -th row equals the **out-degree** $\deg^+(v_i)$.
 - ▶ The column sum of the j -th column equals the **in-degree** $\deg^-(v_j)$.



Examples of adjacency matrices

Undirected case:



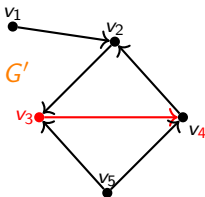
Adjacency matrix of G :

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

► Row sums of the rows: 1, 3, 3, 3, 2.

► $\deg(v_1) = 1, \deg(v_2) = 3, \deg(v_3) = 3, \deg(v_4) = 3, \deg(v_5) = 2$.

Directed case:



Adjacency matrix of G' :

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

► Row sums of the rows: 1, 1, 1, 1, 2.

► $\deg^+(v_1) = 1, \deg^+(v_2) = 1, \deg^+(v_3) = 1, \deg^+(v_4) = 1, \deg^+(v_5) = 2$.

► Column sums of the rows: 0, 2, 2, 2, 0.

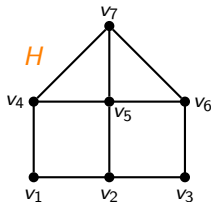
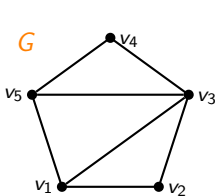
► $\deg^-(v_1) = 0, \deg^-(v_2) = 2, \deg^-(v_3) = 2, \deg^-(v_4) = 2, \deg^-(v_5) = 0$.



Let's be a mathematician for a day!
We have the chance to make the same
discoveries that mathematicians made years ago.



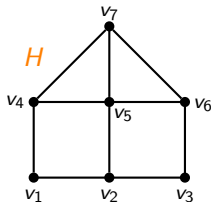
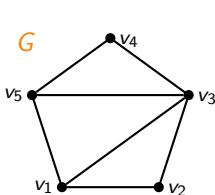
Exercise 1 (20 mins)



There are two simple undirected graphs **G** and **H** depicted above. Let A and B be the adjacency matrices of **G** of **H**, respectively.

1. What is A ? What is B ?
2. How many edges are there in **G**? How about **H**?
3. What is A^2 ? What is B^2 ?
▶ (Hint: How to compute the product of two n -by- n square matrices?)
4. Can you relate the diagonal entries of A^2 to the number of edges in **G**? Similarly, can you relate the diagonal entries of B^2 to the number of edges in **H**? Justify your answer with a proof.

Exercise 1 - Solution



1. [Adjacency matrices A, B of graphs G, H]

Adjacency matrix A :

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Adjacency matrix B :

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

2. [Number of edges in graphs G, H]

► $|E(G)| = 7.$
► $|E(H)| = 10.$

Exercise 1 - Solution (continued)

So far: $|E(G)| = 7$, $|E(H)| = 10$.

3. [Squares of adjacency matrices]

$$A^2 = \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 1 & 2 \\ 2 & 1 & 4 & 1 & 2 \\ 2 & 1 & 1 & 2 & 1 \\ 1 & 2 & 2 & 1 & 3 \end{bmatrix}.$$

$$B^2 = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 1 & 0 & 2 & 0 & 1 \\ 0 & 3 & 0 & 2 & 0 & 2 & 1 \\ 1 & 0 & 2 & 0 & 2 & 0 & 1 \\ 0 & 2 & 0 & 3 & 1 & 2 & 1 \\ 2 & 0 & 2 & 1 & 4 & 1 & 2 \\ 0 & 2 & 0 & 2 & 1 & 3 & 1 \\ 1 & 1 & 1 & 1 & 2 & 1 & 3 \end{bmatrix}.$$

► **Useful Definition:** The **trace** of a square matrix M , denoted by $\text{tr}(M)$, is the sum of the diagonal entries in M .

► **Note:** $\text{tr}(A^2) = 3 + 2 + 4 + 2 + 3 = 14$.

► **Note:** $\text{tr}(B^2) = 2 + 3 + 2 + 3 + 4 + 3 + 3 = 20$.

Observation: $\text{tr}(A^2) = 2|E(G)|$, and $\text{tr}(B^2) = 2|E(H)|$.



Exercise 1 - Solution (continued)

4. [Relate diagonal entries of A^2 to number of edges]

Given a simple undirected graph G with adjacency matrix A , we claim that $\text{tr}(A^2) = 2|E(G)|$.

- ▶ Assume G has n vertices $v_1, \dots, v_n$, and let $A = [a_{i,j}]_{1 \leq i,j \leq n}$.

Idea: What is the (i, i) -th entry of A^2 ?

- ▶ Note: Since G is undirected, the matrix A is symmetric, i.e. $a_{i,j} = a_{j,i}$ for all $1 \leq i, j \leq n$.
- ▶ By matrix multiplication, the (i, i) -th entry of A^2 equals

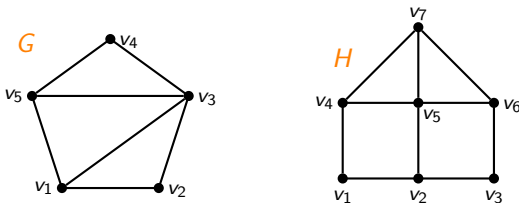
$$a_{i,1}a_{1,i} + a_{i,2}a_{2,i} + \dots + a_{i,n}a_{n,i} = a_{i,1}^2 + a_{i,2}^2 + \dots + a_{i,n}^2,$$

which equals the number of ones in the i -th row of A , since A is a 0/1-matrix, and since $0^2 = 0$, $1^2 = 1$.

- ▶ The number of ones in the i -th row of A is the row sum of the i -th of A , which we know equals $\deg(v_i)$.
- ▶ Thus, $\text{tr}(A^2) = \sum_{i=1}^n \deg(v_i)$.
- ▶ Finally, by the **degree sum formula**, $\sum_{i=1}^n \deg(v_i) = 2|E(G)|$.
- ▶ Therefore, $\text{tr}(A^2) = 2|E(G)|$. □

Challenge of the Day (Try at home)

Definition: Let G be an undirected graph. A **triangle** in G is a vertex-induced subgraph of G isomorphic to K_3 .



We have the same two simple undirected graphs G and H above. Again, let A and B be the adjacency matrices of G of H , respectively.

1. How many triangles are there in G ? How about H ?
2. What is A^3 ? What is B^3 ?
3. Can you relate the diagonal entries of A^3 to the number of triangles in G ? Similarly, can you relate the diagonal entries of B^3 to the number of triangles in H ? Justify your answer with a proof.

This week is bird conservation week!
Do you want to help out with my new
bird conservation initiative?



Actually, in my world, every week is
bird conservation week!



Matching Puzzle [Part I]

It is Day 1 of the bird conservation week.

Our little blue friend John  has six non-bird friends



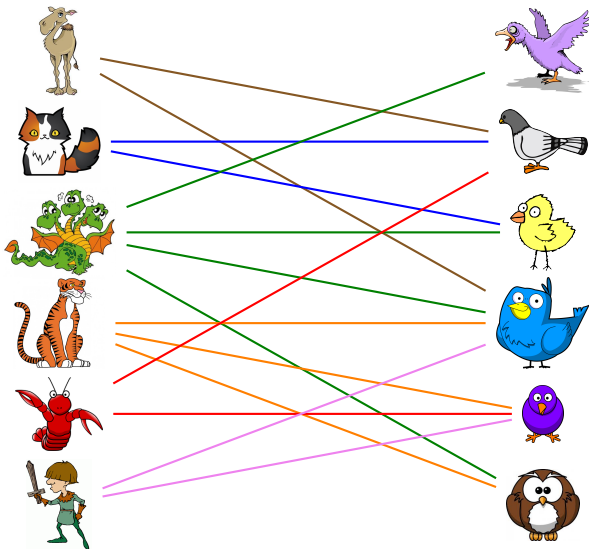
who have volunteered to help with the bird conservation initiative.

- ▶ Every volunteer has to be paired up with a bird to help with conservation efforts.
- ▶ The volunteers have preferences on who they want to work with.

John  has asked for your help to **match up** the volunteers with the birds, so that each volunteer is paired with a bird that he/she/it wants to work with.

Matching Puzzle [Part I] (10 mins)

Here are the preferences of the volunteers.

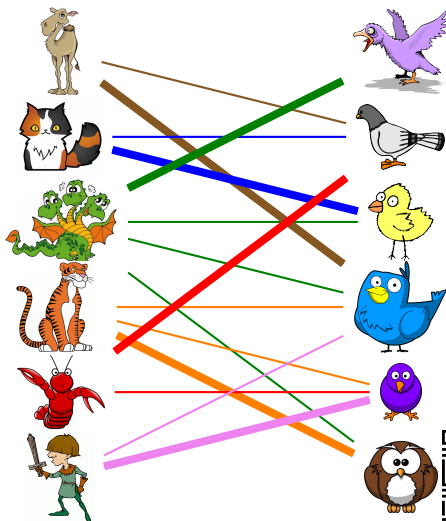
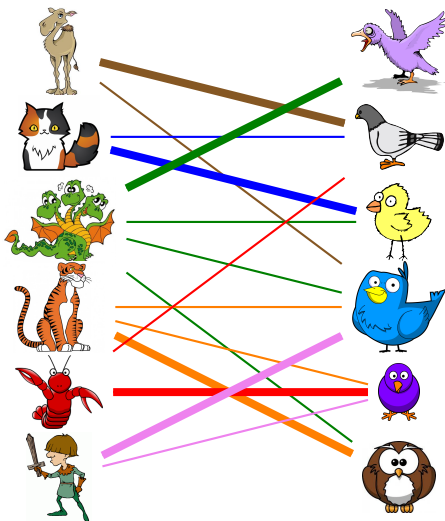


Question: Can you find a matching? In a systematic way?



Matching Puzzle [Part I] - Solution

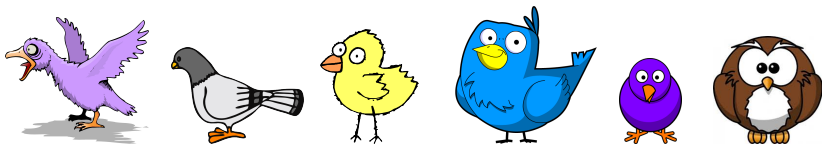
Two possible matchings.



Matching Puzzle [Part II]

It is Day 2 of the bird conservation week.

For Day 2, our six bird friends now want to switch the volunteers they are paired up with.

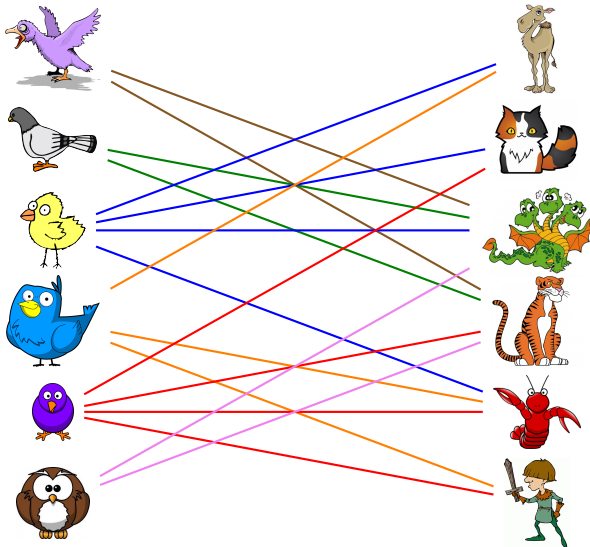


- ▶ Every bird friend has to be paired up with a volunteer.
- ▶ The bird friends have preferences on who they want to work with.

John  has asked for your help to **match up** the birds with the volunteers, so that each bird is paired with a volunteer that he/she/it wants to work with.

Matching Puzzle [Part II] (15 mins)

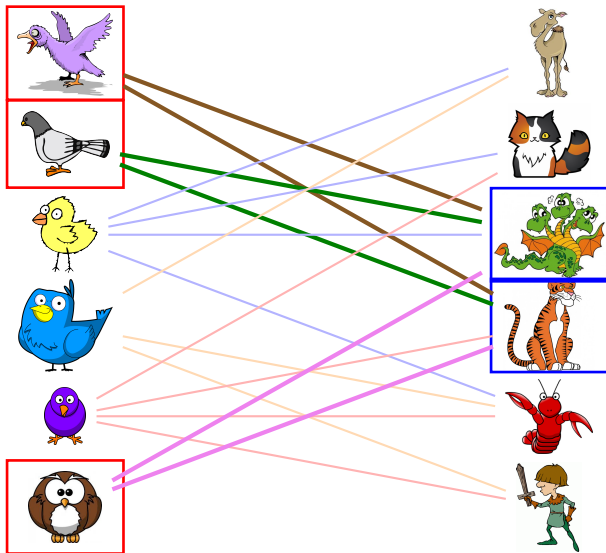
Here are the preferences of our bird friends.



Question: Can you find a matching? Why or why not?



Matching Puzzle [Part II] - Solution



Key Idea: 3 birds    prefer only 2 volunteers  .

► Insufficient volunteers to match with the 3 birds   .

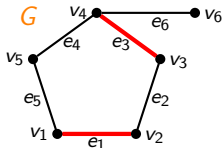
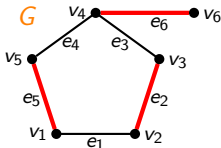


Matchings

Definition: Let $G = (V, E, \phi)$ be an undirected graph with no loops. A **matching** in G is a subset of edges $M \subseteq E$ such that no vertex $v \in V$ is incident to strictly more than one edge in M .

- ▶ A vertex $v \in V$ is called **matched** (w.r.t. M) if $v \in \phi(e)$ for some $e \in E$, and is called **unmatched** or **free** (w.r.t. M) otherwise.
- ▶ A **maximal** matching in G is a matching M such that there is no other possible matching M' in G such that $M \subsetneq M'$.
- ▶ A **maximum** matching in G is a matching M with the largest cardinality over all possible matchings in G .

Example: A graph G with two maximal matchings depicted.



- ▶ Both $M := \{e_2, e_5, e_6\}$ and $M' = \{e_1, e_3\}$ are maximal matchings
- ▶ M is a maximum matching, but M' is not a maximum matching.
- ▶ v_5 is an unmatched (or free) vertex w.r.t. matching M' .



More matching terminology

Definition: Let $G = (V, E, \phi)$ be an undirected graph with no loops. A matching in G is called **perfect** if no vertex is unmatched w.r.t. M .

- ▶ If G has a perfect matching, then $|V|$ must be even, and $|M| = \frac{|V|}{2}$.

Definition: Let $G = (V, E, \phi)$ be an undirected **bipartite** graph with bipartition (V_1, V_2) . A **complete matching from V_1 to V_2** is a matching M such that every $v \in V_1$ is incident to some edge in M .

- ▶ **Note:** A complete matching from V_1 to V_2 is perfect if and only if $|V_1| = |V_2|$.

Example: Two bipartite graphs with complete matchings from V_1 to V_2 depicted.



- ▶ Left depicted matching is a perfect matching.
- ▶ Right depicted matching is not a perfect matching.

Hall's marriage theorem

Let $G = (V, E, \phi)$ be an undirected graph, and let $x \in V$.

- ▶ **Recall:** The **neighborhood** (or **open neighborhood**) of x is the set $N_G(x)$ of all neighbors of x .
 - ▶ i.e. $N_G(x)$ contains the set of all vertices adjacent to x .
- ▶ Given a subset $U \subseteq V$, the **neighborhood** (or **open neighborhood**) of U is the set $N_G(U) := \cup_{x \in U} N_G(x)$.
 - ▶ **Intuition:** $N_G(U)$ contains all vertices adjacent to some vertex in U .
 - ▶ If the context of the graph G is clear, we could simply write $N(U)$.
 - ▶ By default, $N_G(\emptyset)$ is defined to be $\emptyset$.

Theorem: (**Hall's marriage theorem** (1935)) Let G be an undirected **bipartite** graph with bipartition (V_1, V_2) . The following are equivalent:

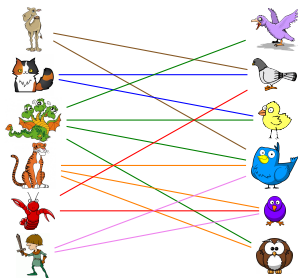
- ▶ There exists a complete matching from V_1 to V_2 .
- ▶ $|N_G(U)| \geq |U|$ for all subsets $U \subseteq V_1$.

This theorem is called “Hall's marriage theorem” because its original formulation was in terms of matching women to men in the “marriage problem”, stated as follows: There are n women and $m \geq n$ men. For each woman, there is a subset of the men, any of whom she will be happy to marry. Any man will be happy to marry. Is it possible to pair up the n women in n marriages?



Another look at the matching puzzle [Part I]

For Day 1 of the bird conservation week: Our task was to find a **complete matching** from {volunteers} to {birds} in the following depicted undirected bipartite graph.



Intuition: To have a complete matching

from $\left\{ \begin{array}{c} \text{cow, cat, frog} \\ \text{cat, dragon, knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird, grey bird, yellow bird} \\ \text{blue bird, purple bird, brown owl} \end{array} \right\}$, we

must have that for any subset U of

$\left\{ \begin{array}{c} \text{cow, cat, frog} \\ \text{cat, dragon, knight} \end{array} \right\}$, the corresponding set of

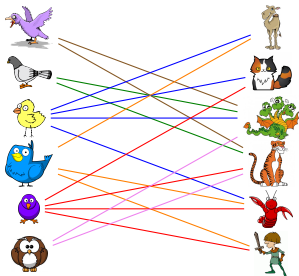
neighbors must have at least $|U|$ birds, otherwise we cannot possibly match the volunteers to the birds.

- ▶ This condition that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow, cat, frog} \\ \text{cat, dragon, knight} \end{array} \right\}$ is a **necessary** condition for there to exist a complete matching.
- ▶ Hall's marriage theorem says that the converse is also true!
- ▶ It means that this condition is a **sufficient** condition as well!
 - ▶ If we want to check whether there exists a complete matching, we only need to check for this condition (i.e. check all subsets).



Another look at the matching puzzle [Part II]

For Day 2 of the bird conservation week: Our task was to find a **complete matching** from {birds} to {volunteers} in the following depicted undirected bipartite graph.



Intuition: To have a complete matching from $\left\{ \begin{array}{c} \text{purple bird, grey bird, yellow bird} \\ \text{blue bird, purple penguin, brown penguin} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{brown dog, black and white cat, green frog} \\ \text{orange cat, red dragon, boy with sword} \end{array} \right\}$, we must have that for any subset U of $\left\{ \begin{array}{c} \text{purple bird, grey bird, yellow bird} \\ \text{blue bird, purple penguin, brown penguin} \end{array} \right\}$, the corresponding set of neighbors must have at least $|U|$ volunteers, otherwise we cannot possibly match the birds to the volunteers.

- ▶ “ $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{purple bird, grey bird, yellow bird} \\ \text{blue bird, purple penguin, brown penguin} \end{array} \right\}$ ” is a **necessary and sufficient** condition for the existence of a complete matching.
- ▶ $2 = |\{ \text{green frog, orange cat} \}| = |N(\{ \text{purple bird, grey bird, brown penguin} \})| < |\{ \text{purple bird, grey bird, brown penguin} \}| = 3$, so Hall's marriage theorem tells us it is not possible to find any complete matching.



Matchings are important

Numerous applications in various areas.

- ▶ Scheduling applications
 - ▶ Exam scheduling, airline scheduling, sports match scheduling, hospital service scheduling, etc.
- ▶ Applications involving matching people or things
 - ▶ Online dating apps, recruitment and job matching, etc.
- ▶ Multimodal data analysis and tracking applications
 - ▶ e.g. match geolocation with other user attributes
 - ▶ e.g. checking for consistency in census records

Note: Matchings are frequently considered in the context of bipartite graphs.

Matching algorithms

Goal: Given an input graph G , returns as output a matching M in G .

- ▶ Variant 1: Return a maximal matching M in G .
- ▶ Variant 2: Return a maximum matching M in G .

Note: Matching algorithms that return maximal matchings as outputs are usually based on greedy algorithms.

- ▶ e.g. find an unmatched vertex v , check the vertices in $N_G(v)$; if there exists an unmatched vertex $u \in N_G(v)$, insert new edge (that is incident to u and v) into the current matching.

Intuition: Maximal matchings could be “very far from” maximum matchings.

- ▶ e.g. For a cycle graph C_{3n} with $3n$ vertices, there exists a maximal matching with n edges, while in contrast, a **maximum** matching of C_{3n} must have $\lfloor \frac{3n}{2} \rfloor$ edges.

Matching algorithms (continued)

Note: Matching algorithms that return maximum matchings are usually based on the following important lemma.

Lemma Let $G = (V, E, \phi)$ be an undirected graph with no loops, and let M be a matching in G (possibly $M = \emptyset$). Let $\ell \in \mathbb{N}$, and let $(v_0, v_1, \dots, v_{2\ell+1})$ be a sequence of pairwise adjacent vertices, such that v_0 and $v_{2\ell+1}$ are **distinct unmatched** vertices, and such that there exist ℓ **distinct** edges $e_1, \dots, e_\ell$ in M satisfying the following:

- ▶ $\phi(e_i) = \{v_{2i-1}, v_{2i}\}$ for every $1 \leq i \leq \ell$ (if $\ell > 0$).

Then there exists $\ell + 1$ distinct edges $e'_0, e'_1, \dots, e'_\ell$ in G satisfying the following:

- ▶ $\phi(e'_i) = \{v_{2i}, v_{2i+1}\}$ for every $0 \leq i \leq \ell$.
- ▶ $(M \setminus \{e_1, \dots, e_\ell\}) \cup \{e'_0, e'_1, \dots, e'_\ell\}$ is a matching in G .

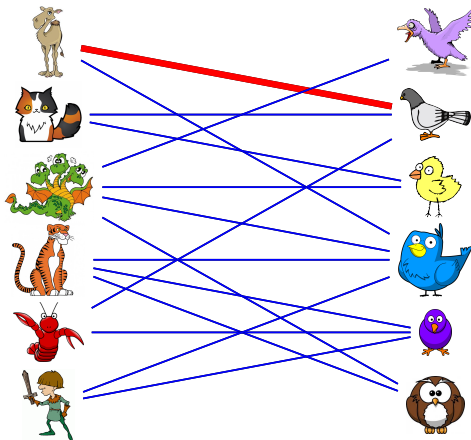
Intuition: We can apply this lemma iteratively to construct matchings of larger and larger cardinalities.

- ▶ We would need to find the distinct unmatched vertices $v_0, v_{2\ell+1}$.
- ▶ We would need to find the edges $e_1, \dots, e_\ell$ from the currently “available” matching.

Intuition for Hall's marriage theorem

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow, cat, frog,} \\ \text{lion, rabbit, knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow, cat, frog,} \\ \text{lion, rabbit, knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird, grey bird,} \\ \text{yellow bird, blue bird,} \\ \text{purple bird, brown owl} \end{array} \right\}$.



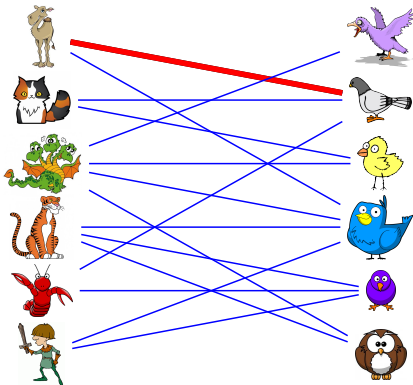
- ▶ Start with any non-empty matching M .
- ▶ Color all edges in M red, and all other edges blue.



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{lion}, \text{insect}, \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{lion}, \text{insect}, \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{parrot}, \text{pigeon}, \text{chicken}, \\ \text{bluebird}, \text{purplebird}, \text{owl} \end{array} \right\}$.



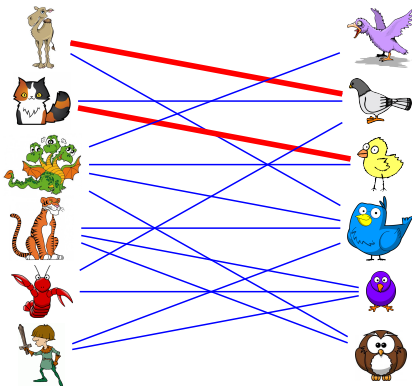
- ▶ If M is not a complete matching, then we can find some unmatched volunteer x , e.g. $x = \text{cat}$.
- ▶ By assumption, $|N(\{x\})| \geq 1$, so we can find a neighbor of x .



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{lion}, \text{insect}, \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{lion}, \text{insect}, \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{parrot}, \text{pigeon}, \text{chicken}, \\ \text{bluebird}, \text{purplebird}, \text{owl} \end{array} \right\}$.



Let $x = \text{cat}$, and let's try to select a neighbor y of x that is not incident to any red edges.

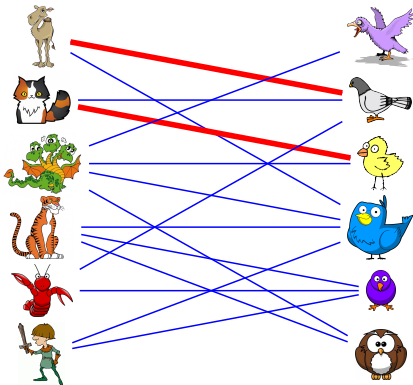
- In our example, we can select $y = \text{chicken}$. Then we can insert a new edge $\{x, y\}$ (i.e. $\{\text{cat}, \text{chicken}\}$) into M , which we color red.



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{tiger}, \text{bug}, \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{tiger}, \text{bug}, \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird}, \text{grey bird}, \text{yellow bird}, \\ \text{blue bird}, \text{purple bird}, \text{brown owl} \end{array} \right\}$.



If M is not a complete matching, then we can find some unmatched volunteer x , e.g. $x = \text{frog}$.

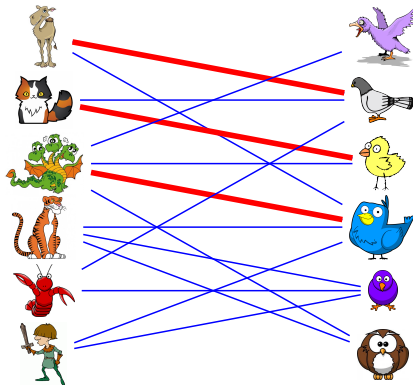
► By assumption, $|N(\{x\})| \geq 1$, so we can find a neighbor of x .



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow} \\ \text{cat} \\ \text{frog} \\ \text{tiger} \\ \text{bug} \\ \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow} \\ \text{cat} \\ \text{frog} \\ \text{tiger} \\ \text{bug} \\ \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird} \\ \text{grey bird} \\ \text{yellow bird} \\ \text{blue bird} \\ \text{purple bird} \\ \text{brown owl} \end{array} \right\}$.



Let $x = \text{frog}$, and let's try to select a neighbor y of x that is not incident to any red edges.

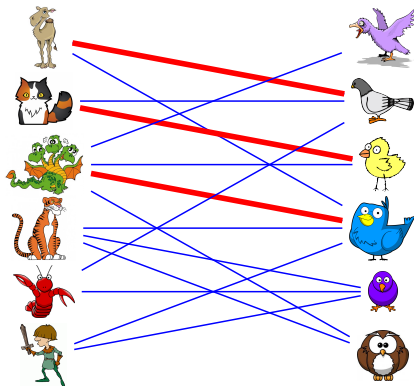
- In our example, we can select $y = \text{blue bird}$. Then we can insert a new edge $\{x, y\}$ (i.e. $\{\text{frog}, \text{blue bird}\}$) into M , which we color red.



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{tiger}, \text{bug}, \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{tiger}, \text{bug}, \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird}, \text{grey bird}, \text{yellow bird}, \\ \text{blue bird}, \text{purple bird}, \text{brown owl} \end{array} \right\}$.



If M is not a complete matching, then we can find some unmatched volunteer x , e.g. $x = \text{tiger}$.

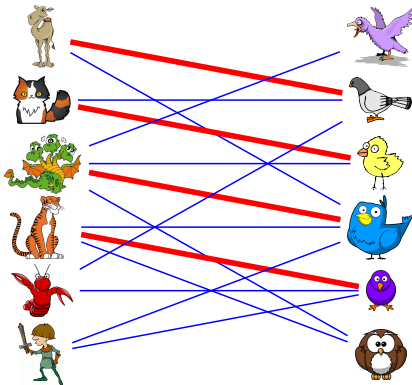
► By assumption, $|N(\{x\})| \geq 1$, so we can find a neighbor of x .



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow, cat, frog, rabbit, knight} \\ \text{cat, rabbit, knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow, cat, frog, rabbit, knight} \\ \text{cat, rabbit, knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird, grey bird, yellow bird} \\ \text{blue bird, purple bird, brown owl} \end{array} \right\}$.



Let $x = \text{cat}$, and let's try to select a neighbor y of x that is not incident to any red edges.

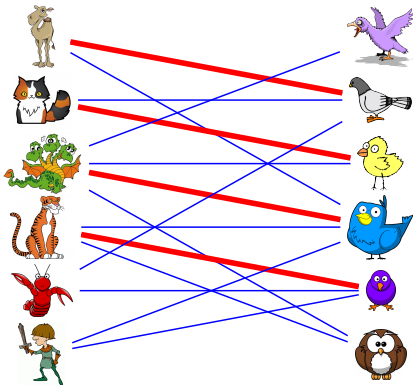
- In our example, we can select $y = \text{purple bird}$. Then we can insert a new edge $\{x, y\}$ (i.e. $\{\text{cat}, \text{purple bird}\}$) into M , which we color red.



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{tiger}, \text{bug}, \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{tiger}, \text{bug}, \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird}, \text{grey bird}, \text{yellow bird}, \\ \text{blue bird}, \text{purple bird}, \text{brown owl} \end{array} \right\}$.



If M is not a complete matching, then we can find some unmatched volunteer x , e.g. $x = \text{bug}$.

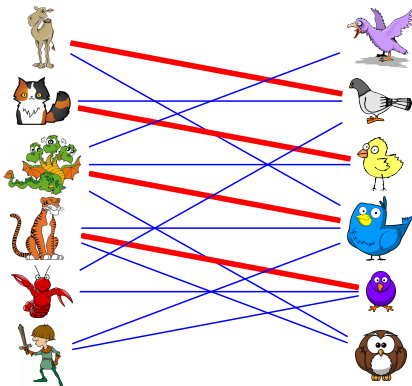
► By assumption, $|N(\{x\})| \geq 1$, so we can find a neighbor of x .



Intuition for Hall's marriage theorem (continued)

Suppose that $|N(U)| \geq |U|$ for all subsets $U \subseteq \left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{bug}, \text{tiger}, \text{knight} \end{array} \right\}$.

Goal: Construct a complete matching from $\left\{ \begin{array}{c} \text{cow}, \text{cat}, \text{frog}, \\ \text{bug}, \text{tiger}, \text{knight} \end{array} \right\}$ to $\left\{ \begin{array}{c} \text{purple bird}, \text{grey bird}, \text{yellow bird}, \\ \text{blue bird}, \text{purple bird}, \text{brown owl} \end{array} \right\}$.

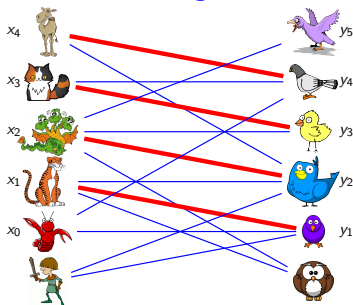


Let $x = \text{bug}$, and let's try to select a neighbor y of x that is not incident to any red edges.

- In our example, every neighbor of x is incident to at least one red edge, i.e. every neighbor of x is a matched bird.



Intuition for Hall's marriage theorem (continued)

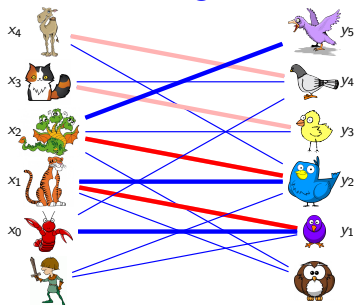


Key Idea 1: Starting with $x_0 = \text{dragon}$, find a bird y_1 incident to x_0 .

- ▶ y_1 is incident to a red edge, so let x_1 be the other endpoint of this red edge.
- ▶ $|N(\{x_0, x_1\})| \geq 2$, so we can find a bird y_2 in $N(\{x_0, x_1\})$ that is not y_1 .
- ▶ y_2 is incident to a red edge, so let x_2 be the other endpoint of this red edge.
- ▶ $|N(\{x_0, x_1, x_2\})| \geq 3$, so we can find a bird y_3 in $N(\{x_0, x_1, x_2\})$ not y_1, y_2 .
- ▶ y_3 is incident to a red edge, so let x_3 be the other endpoint of this red edge.
- ▶ $|N(\{x_0, \dots, x_3\})| \geq 4$, so we can find a bird y_4 in $N(\{x_0, \dots, x_3\})$ not y_1, y_2, y_3 .
- ▶ y_4 is incident to a red edge, so let x_4 be the other endpoint of this red edge.
- ▶ $|N(\{x_0, \dots, x_4\})| \geq 5$, so we can find a bird y_5 in $N(\{x_0, \dots, x_4\})$ not $y_1, \dots, y_4$.
- ▶ Finally, y_5 is NOT incident to a red edge. Stop process.



Intuition for Hall's marriage theorem (continued)

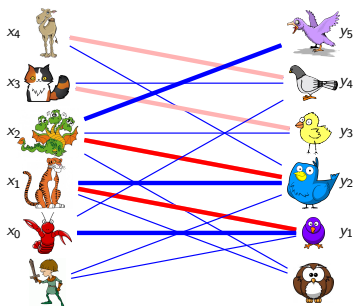


Key Idea 2: Start with the last vertex $y_5 =$  in the sequence.

- ▶ Find some earlier vertex in this sequence adjacent to y_5 , i.e. $x_2 =$ .
- ▶ Define the edge $e_1 := \{x_2, y_5\}$, which is blue.
- ▶ x_2 is adjacent to an earlier vertex in the sequence (because of how we chose x_2). This earlier vertex is $y_2 =$ . Define the edge $e_2 := \{x_2, y_2\}$, which is red.
- ▶ Find some earlier vertex adjacent to y_2 in this sequence, i.e. $x_1 =$ .
- ▶ Define the edge $e_3 := \{x_1, y_2\}$, which is blue.
- ▶ x_1 is adjacent to an earlier vertex in the sequence (because of how we chose x_1). This earlier vertex is $y_1 =$ . Define the edge $e_4 := \{x_1, y_1\}$, which is red.
- ▶ Find some earlier vertex adjacent to y_1 in this sequence, i.e. $x_0 =$ .
- ▶ Define the edge $e_5 := \{x_0, y_1\}$, which is blue.
- ▶ We have reached the first vertex x_0 of the sequence. Stop process.



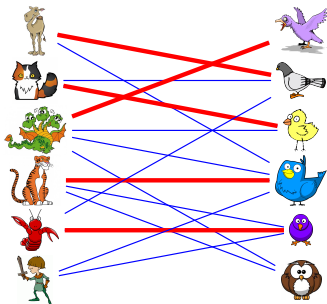
Intuition for Hall's marriage theorem (continued)



Key Idea 3: Look at the sequence of edges $(e_1, e_2, e_2, e_4, e_5)$.

- ▶ 
- ▶ The edges in this sequence alternate in color.
- ▶ This sequence has a total of 3 blue edges and 2 red edges.
- ▶ $|\{\text{blue edges}\}|$ will always be 1 more than $|\{\text{red edges}\}|$.
- ▶ **Idea: Swap red with blue** (and vice versa) in this sequence.
 - ▶ Then there will be 3 red edges and 2 blue edges.

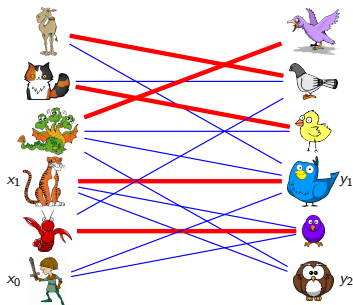
Intuition for Hall's marriage theorem (continued)



After swapping colors, we now have a matching M with 5 (red) edges.

- ▶ M is still not a complete matching, so we can find some unmatched volunteer, e.g. $x =$ .
- ▶ By assumption, $|N(\{x\})| \geq 1$, so we can find a neighbor of x .
- ▶ Let's try to select a neighbor y of x that is not incident to any red edges.
 - ▶ In our example, every neighbor of x is incident to at least one red edge.

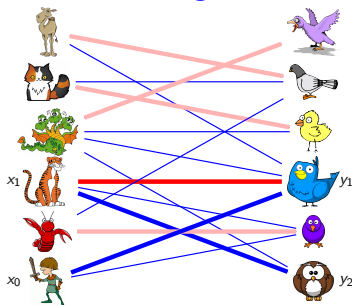
Intuition for Hall's marriage theorem (continued)



Same Key Idea 1: Starting with $x_0 =$ , find a bird y_1 incident to x_0 .

- ▶ y_1 is incident to a red edge, so let x_1 be the other endpoint of this red edge.
- ▶ $|N(\{x_0, x_1\})| \geq 2$, so we can find a bird y_2 in $N(\{x_0, x_1\})$ that is not y_1 .
- ▶ y_2 is NOT incident to a red edge. Stop process.

Intuition for Hall's marriage theorem (continued)

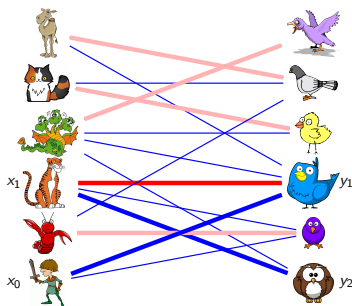


Same Key Idea 2: Start with the last vertex $y_2 = \text{owl}$ in the sequence (x_0, y_1, x_1, y_2) .

- ▶ Find some earlier vertex in this sequence adjacent to y_2 , i.e. $x_1 = \text{tiger}$.
- ▶ Define the edge $e_1 := \{x_1, y_2\}$, which is blue.
- ▶ x_1 is adjacent to an earlier vertex in the sequence (because of how we chose x_1). This earlier vertex is $y_1 = \text{blue bird}$. Define the edge $e_2 := \{x_1, y_1\}$, which is red.
- ▶ Find some earlier vertex in this sequence adjacent to y_1 , i.e. $x_0 = \text{knight}$.
- ▶ Define the edge $e_3 := \{x_0, y_1\}$, which is blue.
- ▶ We have reached the first vertex x_0 of the sequence. Stop process.



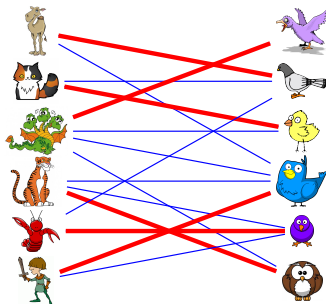
Intuition for Hall's marriage theorem (continued)



Same Key Idea 3: Look at the sequence of edges (e_1, e_2, e_3) .

- ▶ 
- ▶ The edges in this sequence alternate in color.
- ▶ This sequence has a total of 2 blue edges and 1 red edge.
- ▶ $|\{\text{blue edges}\}|$ will always be 1 more than $|\{\text{red edges}\}|$.
- ▶ **Idea: Swap red with blue** (and vice versa) in this sequence.
 - ▶ Then there will be 2 red edges and 1 blue edges.

Intuition for Hall's marriage theorem (continued)



After swapping colors, we now have a matching M with 6 (red) edges.

- M is now a complete matching, so we have explicitly constructed our complete matching. $\square$

Summary

- ▶ Exercise on adjacency matrices
- ▶ Class activity: Matching puzzle
- ▶ Hall's marriage theorem
- ▶ Discussion on matching algorithms